Prithu Paresh Das

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OVERVIEW

Hands-on experience in designing, fabricating and building auxetic structures, shape-memory actuators, and robotic-manipulation hardware in both metals and polymers. Incoming Mechanical Engineering MSE student at the University of Michigan (Fall 2026).

Please do click the link beside each project to see its models and drawings!

Interactive 3D models and build videos for every project below are available on the portfolio: sawbl4de.github.io

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani — Hyderabad, India
Bachelor of Engineering in Mechanical Engineering & Minor in Data Science (GPA: 8.14/10, MGPA: 8.72/10) — Oct '22 - May '26

- **Relevant Coursework:** Computer Aided Design, Design of Machine Elements, Advanced Manufacturing Processes, Manufacturing Management, Advanced Mechanics of Solids
- **Teaching Assistant:** Advanced Mechanics of Solids; Vibrations & Control

HARDWARE DESIGN AND FABRICATION

Auxetic Structures - sawbl4de.github.io/auxetics

- Designed and built four lattice geometries end-to-end from CAD and finite-element analysis to precision laser cutting - in stainless steel, mild steel, and NiTiNOL shape-memory alloy.
- Tuned the laser cutting process to control heat damage in NiTiNOL, a material that is difficult to machine cleanly.
- Built a shaker test rig with paired accelerometers and measured roughly a **40% drop** in transmitted vibration when compared with solid plates.
- Designed a custom fixture to hold samples square in the tensile tester; ran tensile tests at varied speeds and temperatures and prepared samples for DSC, XRD, and SEM.
- Work led to **two conference papers**, a submitted **journal manuscript**, and an **Indian patent application**.

Shape-Memory Alloy Actuators - sawbl4de.github.io/sma

- Designed a motor-free actuator that opens and closes louvre flaps using heated NiTiNOL springs.
- Wound custom springs by hand from 1 mm diameter steel and NiTiNOL wire, then heat-set them in a furnace at 450·C - demanding work at that wire thickness.
- Fabricated the full assembly myself: laser-cut guides, sheared and drilled flaps, and a turned-and-ground sliding shaft.

Single-Stage Gearbox - sawbl4de.github.io/gearbox

- Designed a complete single-stage gearbox in Fusion 360 and machined every part from mild-steel and cast-iron.
- Turned the blanks on a lathe, cut the gear teeth, and machined a housing that holds the shafts aligned to press-fit tolerances.
- Assembled and ran it under load — smooth power transfer with no binding.
- Developed a Digital Twin of the gearbox for real-time condition-monitoring and control.

SCARA Robot & 3D-Printed Manipulation Hardware - sawbl4de.github.io/scara1

- Designed and 3D-printed a precision end-effector and sample holder for micron-scale material handling as part of the research SCARA setup at OUSUMS Lab, IISc Bangalore.
- Developed and 3D-printed custom parts (ABS and PLA) for manipulation experiments in my thesis work.
- Designed and built a working 3-axis SCARA robot from PLA and acrylic - base, links, joints, and end-effector.

ROBOTICS RESEARCH & OTHER EXPERIENCES

Indian Institute of Science (IISc), Bangalore — Bengaluru, Karnataka
Research Intern - Robotics and Autonomous Systems — June '25 - Present

Optical Ultrafast Spectroscopy and Understanding Material Synthesis (OUSUMS) Lab

- Supervised by Prof. Akshay Singh, Department of Physics.
- Developed an **image-based visual servoing** pipeline for a Mitsubishi SCARA to enable **micron-level manipulation** and stacking of 2D materials to build heterostructures.
- Replaced open-loop motion with continuous IBVS feedback, eliminating cumulative image-space tracking error and achieving **1–2 micron accuracy**.
- Reduced transfer time from **20 minutes to 2 minutes** and executed the first **semi-automated heterostructure assembly**.

Data Augmented Control of Autonomous Systems (DACAS) Lab

- Supervised by Prof. Jishnu Keshavan, Department of Mechanical Engineering.
- Performed dynamic model identification and implemented an **inverse-dynamics torque controller** on a 6-DOF PiPER manipulator, enabling model-consistent torque commands.
- Identified failures due to modeling errors, **reconstructed and validated** the robot's kinematic and dynamic model by correcting joint axes, inertial properties, and frame conventions in the URDF.

- Developed safe stiffness-damping profiles for manipulation under dynamic coupling and external disturbances.
- Performed joint-level impedance characterization under varying loads, revealing physical interaction limits necessary for safe torque control.
- Developing a real-time image-based visual servoing framework for gesture-based control and ball-catching tasks.

Safety, Reliability And Maintainability Centre, CEMILAC Bengaluru, Karnataka
Research Intern - Reliability, Safety and Airworthiness Analysis May '24 - July '24

- Undertook an academic research project at the premier national aerospace laboratory responsible for airworthiness certification and safety evaluation.
- Developed a novel probabilistic method to estimate the **Reliability** and **Unavailability of One-Shot Applications**.
- Validated the developed method for a Payload Delivery System using ISOGRAPH and Monte Carlo simulations.
- Executed full-scale **System Safety Analysis**, **FMECA, PHA and Risk Assessment** on the **Landing Gear** unit of an aircraft & other aviation hardware under Directorate General of Civil Aviation standards.

PUBLICATIONS AND PATENTS

Conference Papers

- *Damping Characteristics of Auxetic Structures Made of Shape Memory Alloys - 20th International Conference on Vibration Engineering & Technology Of Machinery, IIT Guwahati Dec 2025* (Accepted)
- *Estimating the Vibration Attenuation Characteristics of Re-Entrant Bow-Tie Shaped Auxetic Structures through Experimental and Computational Analysis - 13th International Conference on Precision, Meso, Micro and Nano Engineering, NIT Calicut Dec 2024* (Presented)

Journal Paper (Submitted)

NiTi-Based Auxetic Structures for Enhanced Vibration Damping: Design, Mechanics, and Performance Evaluation – Submitted to Smart Materials and Structures

Patent

Contributed to Indian Patent Application No. 202311076100: "Shape Memory Alloy Based Auxetic Structure for Damping Applications"

TECHNICAL SKILLS

- **Design/CAD/FEA:** AutoCAD, Creo, Fusion 360; finite-element analysis with Ansys and COMSOL
- **Fabrication:** Laser cutting, 3D Printing, CNC, lathe, milling, gear cutting, drilling, welding, surface grinding & polishing, furnace heat-treatment
- **Materials:** Stainless steel (304 / 316), mild steel, cast iron, NiTiNOL (shape-memory alloy), acrylic
- **Testing:** Tensile testing (UTM), shaker / vibration testing, sample preparation for DSC, XRD, and SEM
- **Robotics and Software:** Python, TensorFlow, ROS 2 (Beginner), Gazebo, Pinocchio; LaTeX, Git

WORKSHOPS AND CERTIFICATIONS

ADAS/Intelligent Systems Workshop - Dept. of Mechanical Engineering, BITS Pilani; WILP, BITS Pilani Nov '24

- Gained industry insights into ADAS and intelligent control systems, focusing on automation, sensor fusion, and autonomy.

Supervised Machine Learning: Regression and Classification Oct '24

- Stanford Online, DeepLearning.AI - Credential ID D24TG6FFZHTR

Navigating the Pathways of Publishing in High-Quality Journals Aug '24

- Gained insights into research workflow, journal selection, and publishing strategies to improve article acceptance.

AWARDS AND ACHIEVEMENTS

- Raising A Mathematician Training Program - Selected among **top 100** students of the country to be trained in Advanced Mathematics at a residential camp in Mumbai - May '18
- 1st Dan Black Belt in Okinawa Shorin-Ryu Shorin-Kan Karate, with **12+ years** of practice. Trained and prepared over 50 students for regional and state tournaments.